

ARM® Mali™ 3D Graphics System Solution

Ashley Stevens – December 2006



Introduction

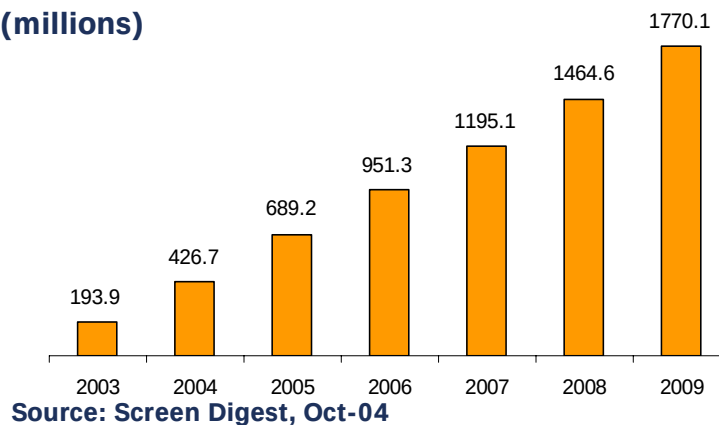
Today's mobile and consumer devices feature complex graphical user interfaces, and many incorporate gaming capabilities. This has resulted in increased capability and performance requirements, whilst requiring low power consumption and low cost. The Mali™ family of graphics IP cores from ARM® is designed to meet the market requirements for high-performance 2D and 3D graphics and video decode and encode in mobile devices. ARM Mali products are incorporated into end-user devices by system integrators and System-on-Chip (SoC) vendors serving mobile phone and PDA manufacturers, mobile game console vendors and manufacturers of infotainment systems for the automotive industry; all areas that require sophisticated graphics capabilities.

This paper describes the system integration requirements for a Mali core, using as example an ARM1176JZF-S™ based applications processor that has a Mali200 Pixel Processor and MaliGP2 Geometry Processor included within the system design.

Market

With voice revenue falling mobile operators are seeking new products that generate increased revenues and help to increase ARPU (Average Revenue Per User).

Games-Enabled Handset Installed Base (millions)



Source: Screen Digest, Oct-04

The global market for mobile video games is predicted to increase from a little under \$1Bn in 2004 to \$4.7Bn by 2010¹ and \$7.2bn in 2011². Including advanced graphics capability into a new applications processor is now

¹ Source: Strategy Analytics December 2005

² Source : Informa Telecom August 2006

common to ensure that a new hand-held device has the correct feature-set to enable content providers and operators to maximise their revenue-generating opportunities.



The Mali 200 Pixel Processor and MaliGP2 Geometry Processor

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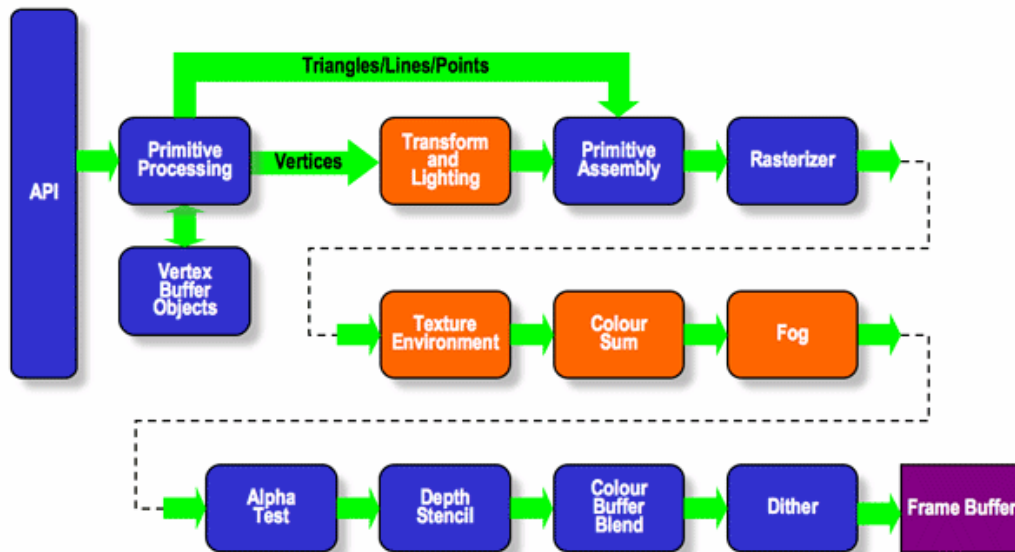
3D graphics accelerators traditionally require very high memory bandwidth. The Mali family minimises memory bandwidth by using a tile-based rendering scheme. Tile-based renderers are also known as ‘deferred renderers’ because rendering is deferred until a complete frame of polygons (triangles) is ready to be rendered. Tile-based schemes break the screen into tiles and render a tile at a time to an on-chip tile-memory, meaning that many memory accesses that would go off-chip with a traditional renderer remain on-chip using Mali cores. Since off-chip memory accesses are roughly an order of magnitude higher power than on-chip memory accesses, this also helps minimise power consumption – key for a mobile consumer device which requires good battery life.

A drawback of tile-based renderers is cost – the additional complexity compared with simple ‘immediate mode’ renderers results in high gate counts and therefore additional silicon area and hence cost. The Mali family uses a unique blend of tile-based rendering and traditional ‘immediate-mode’ rendering, thus simultaneously minimising memory bandwidth **and** cost/area, enabling Mali cores to be significantly smaller than competing mobile tile-based renderers.

Together the Mali200 and MaliGP2 cores implement both the OpenGL ES 1.1 API **and** the OpenGL ES 2.0 programmable API. The OpenGL ES 2.0 API provides flexibility and high-quality via a programmable rendering pipeline which replaces the fixed-function pipeline in OpenGL ES 1.1. This enables very high quality at with low numbers of polygons – minimizing memory bandwidth whilst maximizing image quality. The Mali200 core implements the OpenGL ES 2.0 pipeline stages from the rasterizer onwards. The MaliGP2 implements the Vertex Shader part of the graphics pipeline. As well as

OpenGL ES, the Mali cores can also implement other APIs such as Microsoft® DirectX Mobile.

OpenGL ES 1.x Fixed Function Pipeline



OpenGL ES 2.0 Programmable Pipeline

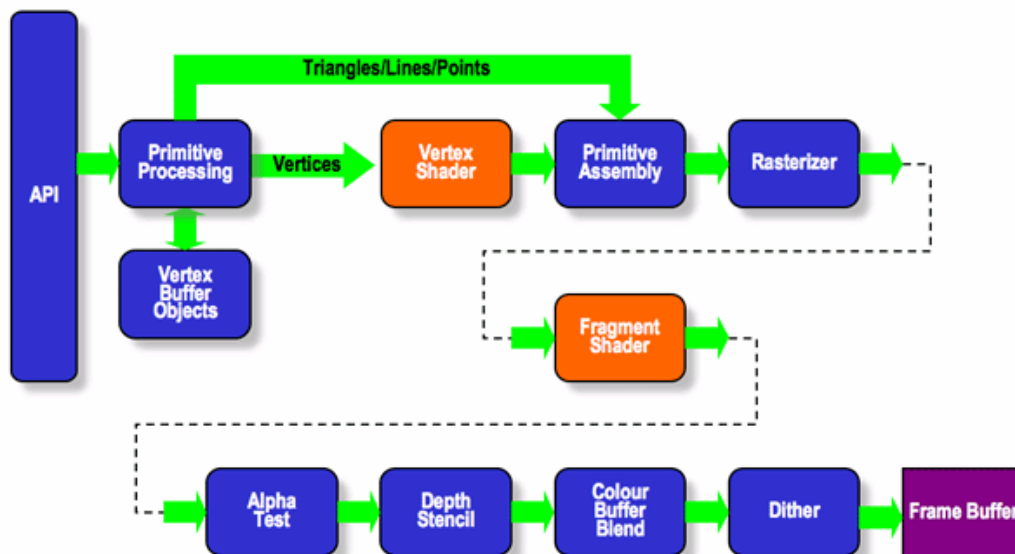


Image quality

To make the most of the small displays of mobile phones, PDAs and mobile GPS devices, Mali cores offer Full Scene Anti-Aliasing (FSAA). FSAA reduces jaggies and improves image quality. Mali cores use a patent-pending

technique that enables 4X anti-aliasing at virtually no cost to memory bandwidth and frame-rate, whilst 16X anti-aliasing can be performed albeit with some increase in bandwidth and reduction in frame rate achieved.



Mali200 also supports High Dynamic Range (HDR) rendering which uses floating-point values to represent the huge range of light values that can occur in natural scenes. This results in more life-like and striking images as shown in the example below.



Mali200 HDR Rendering Demo

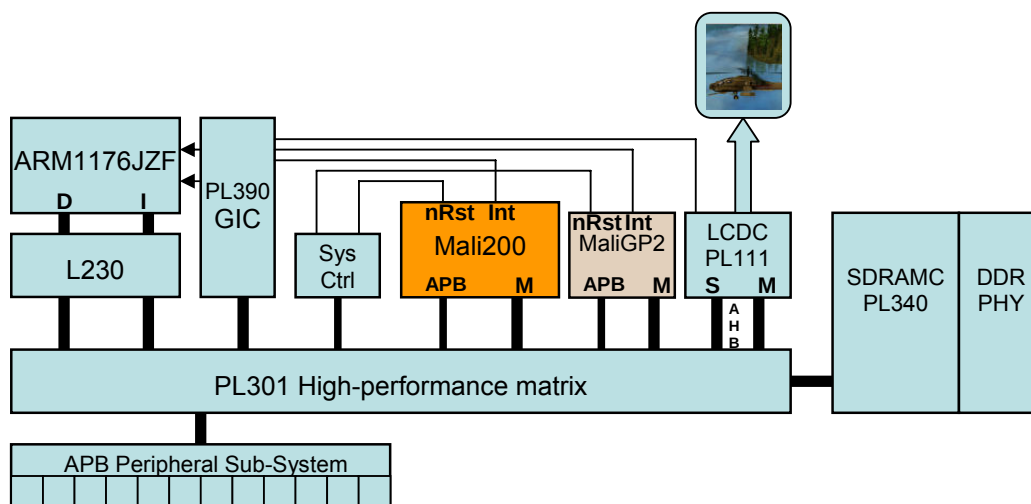
High dynamic range cubemap reflections
with trilinear floating point filtering

System Integration

The Mali graphics cores are designed for a Unified Memory Architecture where a single memory sub-system is shared between the main processor and the graphics accelerator. The Mali core's primary system interface is an AMBA 3 AXI master interface. An AMBA 2 APB™ interface provides for programming of on-chip registers. Accesses to Mali registers are minimized because all graphics processing commands for each frame are assembled in memory with little more than the base address of the commands in memory needing to be programmed into registers. In addition to the AMBA 3 AXI and AMBA 2 APB interfaces each core also has a core clock input, to clock the Mali core, a bus clock input to clock the bus interface logic, a reset input and an interrupt output. The reset input should be under software control of the processor, and the interrupt output is routed to the system interrupt controller such as the ARM PrimeCell® PL390 Generic Interrupt Controller (GIC). The resulting video frame is written by the Mali200 pixel processor core to main memory where it's accessed by the display controller.

Data flow

All 3D objects to be displayed are broken down into lists of triangles, the triangles being made up of 3 vertices, the vertices being points in X,Y,Z space. Assuming the MaliGP2 geometry processor is present, the vertices and commands for the MaliGP2 are written by the ARM processor to memory. The MaliGP2 processes the commands and vertices and writes the transformed vertices back to main memory. The Mali200 then rasterizes the triangles – it reads vertices and commands from main memory and renders a tile at a time. The tiles are then written to memory to be read and displayed by an LCD or VGA controller. As the Mali cores are deferred renderers, the pipeline between the ARM core and the MaliGP2 and between the MaliGP2 and the Mali200 is on a frame basis – a complete frame of data is generated and stored in main memory between each component in the pipeline flow.



Typical System Integration with ARM1176JZF-S

3D Graphics Memory Access Profile

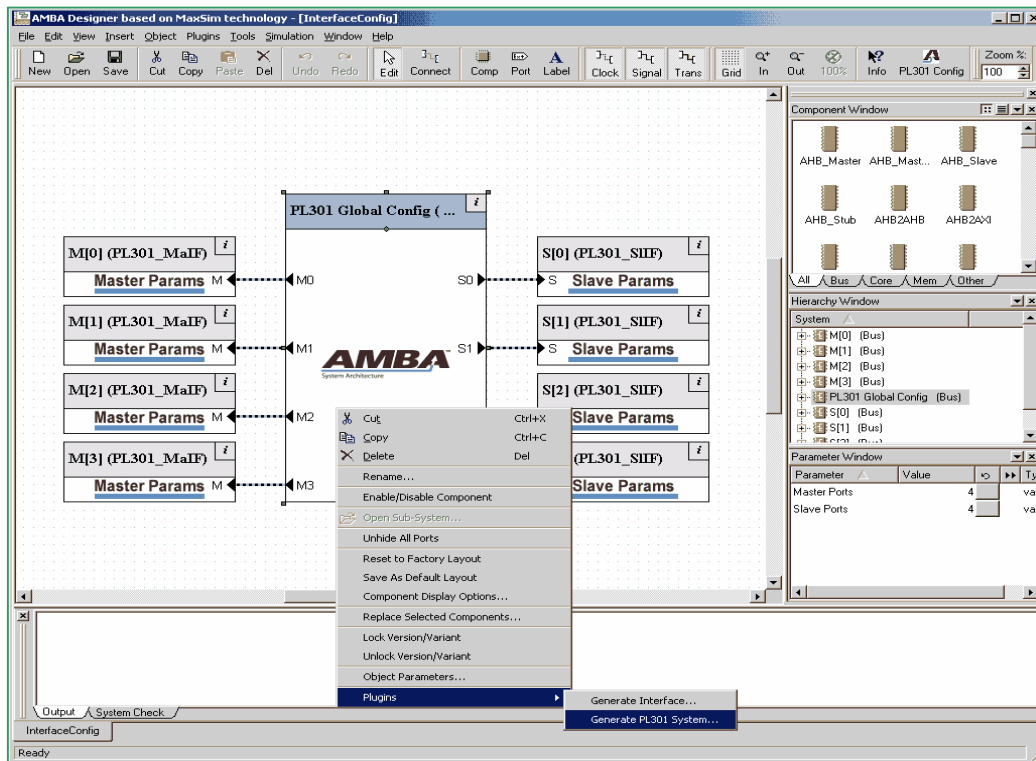
The Mali graphics cores require less memory bandwidth than competitive solutions, but memory system bandwidth is still important. The Mali cores are designed to share SDRAM with the main processor. Today's SDRAM devices have high clock speeds and high theoretical maximum bandwidth, but long latency. The Mali cores are designed to work with SDRAM. The transactions from Mali cores are mostly bursts of 4 or 8 words. 3D graphics engines differ from general-purpose processors in that there are few dependencies between individual memory accesses. A memory transaction for another pixel can be issued before data is returned for the previous pixel as there is no dependency between them. Thus the 3D core can issue multiple outstanding transactions.

AMBA® 3 AXI™ Interface Protocol

ARM AMBA® 3 AXI™ is ideally suited to graphics cores as it supports multiple outstanding transactions. A master issuing a transaction can tag the transaction with a unique ID. Each unique ID is akin to a separate virtual master. Slaves supporting multiple outstanding transactions can accept multiple addresses and even return read data out of order. The combination of a master like Mali200 that can issue multiple outstanding transactions over AMBA 3 AXI with an SDRAM controller that can accept multiple outstanding transactions enables deep pipelining of transactions to hide the long latency of SDRAM accesses, maximising actual memory bandwidth.

ARM PrimeCell® PL301 High-Performance Matrix

The ARM PrimeCell PL301 High-Performance Matrix implements an AMBA 3 AXI-based interconnect. It supports both AXI and AHB™ master and slaves and APB slaves. It supports a range of master and slave bus widths – 64 bit, 32 bit and 128-bit. Most ARM1176JZ-S based systems use a 64-bit interconnect as this is the width of the AXI interfaces on the processor core. The interconnect matrix may be fully or sparsely connected. A quality-of-service (QoS) scheme ensures that latency-critical masters can always request data from a QoS-supporting slave, such as the ARM PrimeCell PL340 SDRAM controller (or PL341 DDR2 SDRAM controller). As the PL301 is highly configurable, it is delivered within the ARM AMBA Designer tool, a graphical user interface (GUI) that enables the customer to easily configure and simulate a custom interconnect for their system.



AMBA Designer GUI enables PL301 interconnect configuration

ARM PrimeCell SDRAM Controllers

The ARM PrimeCell PL340 SDRAM controller supports SDR and DDR memory, including mobile and low-power variants of each over 16, 32 and 64 bit buses, with clock rates up to 166MHz. The number of outstanding transactions supported is configurable – typically eight would be supported. A programmable quality-of-service (QoS) scheme ensures latency-critical masters such as LCD or video controllers meet their latency requirements. The Mali cores however are not latency-critical.

A typical mobile application would use 16 or 32-bit mobile DDR clocked at between 100 and 166MHz (200 to 333MHz data rate). For higher performance systems, the ARM PrimeCell PL341 DDR2 SDRAM controller supports DDR2 clock speeds up to 400MHz (800MHz data rate). The PL341 supports more outstanding transactions – although configurable a typical application would support twelve. To complete the solution ARM offers (via its Artisan physical IP division) DDR and DDR2 pads and the PLL and DLLs needed to implement the complete DDR pad interface (PHY).

ARM PrimeCell PL111 Color LCD/VGA Controller

The Mali rendering engines render graphics to a frame buffer in memory to be displayed by an LCD or video controller. The ARM PrimeCell PL111 Color LCD controller supports monitors and LCD panels up to 1024x1024 resolution. In addition to active TFT LCDs it also supports older single and dual-panel STN LCD displays. The PL111 LCD controller is a 64-bit AMBA AHB bus master. When generating the PL301 interconnect, the AMBA Designer tool

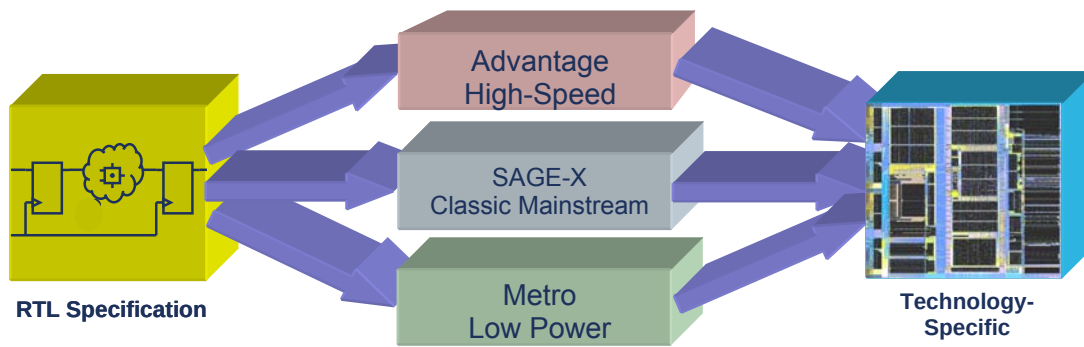
automatically includes the AHB to AXI bridge needed to interface the PL111 to the AXI interconnect matrix and to the PL340 memory controller. The QoS facility in the PL340 and PL301 is ideally suited to low-latency masters such as the PL111 LCD controller. The PL111 supports frame buffer formats of RGB 5:6:5, ARGB 1:5:5:5, 4:4:4:4 and 8:8:8:8, all of which are also supported by the Mali200 core, as well as some others not supported by the ARM LCD controller.

Silicon Implementation

Although Mali cores are smaller than competitive solutions, 3D graphics is complex and the Mali cores still comprise a large number of gates. Mali solutions range from 290K gates for the Mali55 to approximately 1,340K³ gates for the Mali200 with MaliGP2. Fortunately Mali cores provide excellent performance at modest clock speeds. A typical ARM1176JZ-S based SoC would run the CPU between 300 and 600MHz, and the AMBA interconnect at either $\frac{1}{2}$ or $\frac{1}{3}$ of the CPU clock. For example, the interconnect could run at 133MHz and the CPU at 400MHz or 533MHz; or the interconnect at 166MHz and the CPU at 500MHz. (Of course, if ARM Intelligent Energy Management - IEM is used then the CPU clock could be asynchronous to the bus and dynamic). The Mali cores are designed for moderate clock rates whilst achieving high performance at minimal power consumption. They therefore typically run at the interconnect clock frequency (100 to 166MHz). For systems where this is not practical the Mali SoC interface may be asynchronous.

The Mali cores are delivered as synthesizable Verilog and use standard compiled RAMs. To further minimise power and area, a low-power, low-area library such as the ARM Artisan Metro library may be targeted. The Metro library enables low-power and low-area implementations for designs not requiring the highest clock rates. For very high performance implementations where power and area are less of a concern, for example in tethered consumer electronics applications like set-top boxes (STB) then the ARM Artisan Advantage libraries can be used. For mid-level performance on older technology nodes (eg 130nm) the SAGE-X library can be used. For many foundries, for example the IBM/Chartered/Samsung consortium, ARM Artisan library IP is free to the end-user and can be downloaded from the ARM website at : <http://www.arm.com/products/physicalip/productsservices.html>

³ Gates counts based on NAND2X1 (2-input NAND gates with drive strength 1)



ARM libraries – Metro low-power, Sage-X mid-range, Advantage high-performance

Instantiated Memories

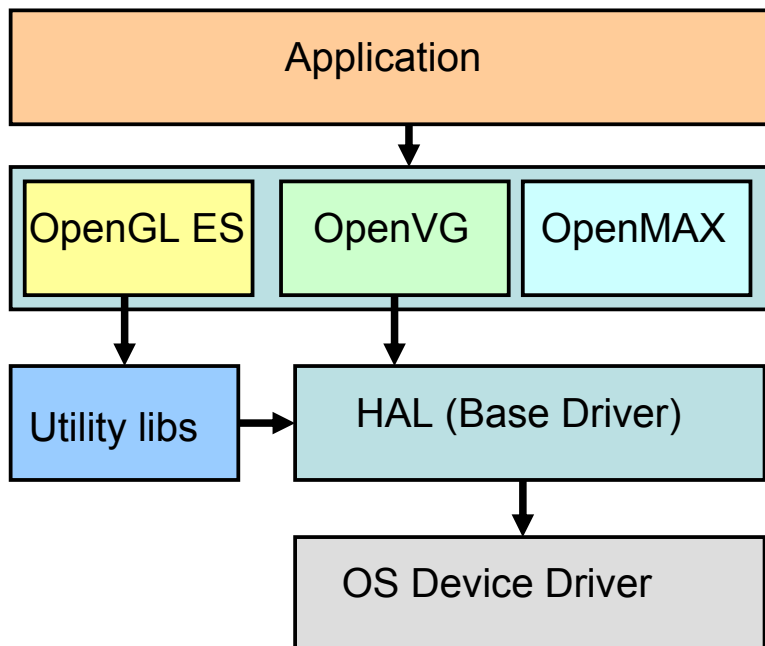
The Mali200 core makes extensive use of compiled memories. There are 32 compiled RAMs in a typical Mali200 configuration making a total of 35KB of RAM. In addition the MaliGP2 has 13KB of RAM in multiple instances. Some of those instances may in fact be implemented as several RAMs, depending on the RAM library and the RAM sizes available. Smaller RAMs may be implemented as register files. The cross-over point between when to use a RAM or a register-file depends on the RAM bit-count and library targeted. ARM Artisan RAM compilers and register file compilers can be used to implement the RAMs.

Multiplier Compilation/Synthesis

The Mali cores make extensive use of synthesized/compiled multipliers. It is important, in minimizing the area of the core while maintaining performance, that a properly optimized gate-level compiler be used, eg the Synopsys DesignWare library. The ARM standard cell families – SAGE, Metro, and Advantage have special-purpose arithmetic cells optimized for use within DesignWare.

Software Drivers

The Mali200 and MaliGP2 cores are supplied with drivers for OpenGL ES2.0, OpenGL ES1.1 (for compatibility with the fixed-function API) and OpenVG Vector Graphics APIs. Licensees receive drivers for **one** OS of their choice (chosen from Linux, Symbian, Windows Mobile) and may optionally license (at additional cost) the driver-sets for additional operating systems. In addition to OpenGL ES, the Microsoft DirectX Mobile API will be supported on Microsoft Windows Mobile. The OpenMAX IL 1.0 API will also be supported for video acceleration.



Mali Software Stack – Only the OS device driver is linked to the kernel

Summary

The ARM Mali200 pixel processor and MaliGP2 geometry processor provide the basis of a complete 2D and 3D graphics for today's mobile handheld device. The low memory-bandwidth of the Mali solution makes it especially appropriate for mobile applications. But there's more than just a 3D core to make a complete graphics solution. ARM offers a complete system solution – the software drivers, the processor on which they run, the AMBA 3 AXI interconnect, high-throughput SDRAM controllers, and a LCD/VGA video controller. ARM also offers physical IP (standard cell libraries and memories) to implement the cores. Implementers of Mali cores can choose libraries to suit their application. For mobile, battery-powered applications the ARM Artisan Metro libraries offer high-density to minimise cost and low-power to maximise battery-life. Together these products provide a complete low-power, high-performance 2D and 3D graphics solution for today's advanced mobile and embedded devices.

About the ARM Mali family of graphics accelerators

The ARM Mali family consists of the Mali55 pixel rendering engine which provides advanced 3D and 3D graphics acceleration at very low cost, marking it ideal for low-end and mid-range mobile phones; and the Mali200/MaliGP2 which provide high-end programmable graphics on mobile and consumer devices. Mali solutions span a full range of gate-count, performance and features.

Mali family cores

	Mali200	MaliGP2	Mali55
Anti-Aliasing	4X / 16X	4X / 16X	4X / 16X
OpenGL®ES 1.x	√	√	√
OpenGL®ES 2.x	√	√	x
OpenVG 1.x	√	n/a	√
Gate count (logic)	1100K	240K	290K
SRAM	35kB	13kB	5kB
Die Area	5.5mm ²	1.5mm ²	1.1 mm ²
Power	0.7 mW/MHz	0.3mW/MHz	0.25mW/MHz
Max CLK	250MHz	250MHz	200MHz
Fill rate / s	250Mpix/s	NA	100Mpix/s
Triangles / s	8M	8M	1M
DirectX9 Vista extensions	√	√	x

Rendering and Computation Equivalence Performance Notes

- Performance figures based on maximum clock rate
- 4X FSAA turned on (1 pix = 4 sub-samples) and bi-linear filtering
- Fill rate and geometry expressed as max theoretical performance

Area, Gate Count & Power Notes

- ARM Advantage standard cell library, process TSMC CLN90G
- Gate counts expressed as Nand2X1AD, includes scan
- Power based on 90nmG process running scene with Overdraw=4,30fps, 4X FSAA, bi-linear filtering

About OpenGL ES and OpenVG

OpenGL®ES and OpenVG™ are royalty-free cross-platform APIs maintained and promoted by the Khronos™ Group. The Khronos Group is a member-funded industry consortium focused on the creation of open-standard, royalty-free APIs to enable the authoring and accelerated playback of dynamic media on a wide variety of platforms and devices.

OpenGL ES is a royalty-free, cross-platform API for full-function 2D and 3D graphics on embedded systems - including consoles, phones, appliances and vehicles. It consists of well-defined subsets of desktop OpenGL, creating a flexible and powerful low-level interface between software and graphics acceleration. OpenGL ES includes profiles for floating-point and fixed-point systems and the EGL™ specification for portably binding to native windowing systems. OpenGL ES 1.X is for fixed function hardware and offers acceleration, image quality and performance. OpenGL ES 2.X enables fully programmable 3D graphics.

OpenVG is a royalty-free, cross-platform API that provides a low-level hardware acceleration interface for vector graphics libraries such as Flash and SVG. OpenVG is targeted primarily at handheld devices that require portable acceleration of high-quality vector graphics for compelling user interfaces and text on small screen devices - while enabling hardware acceleration to provide fluidly interactive performance at very low power levels.